

Bases de Datos 2

Teoría 05

Federico Orlando

BBDD Relacionales

Limitaciones



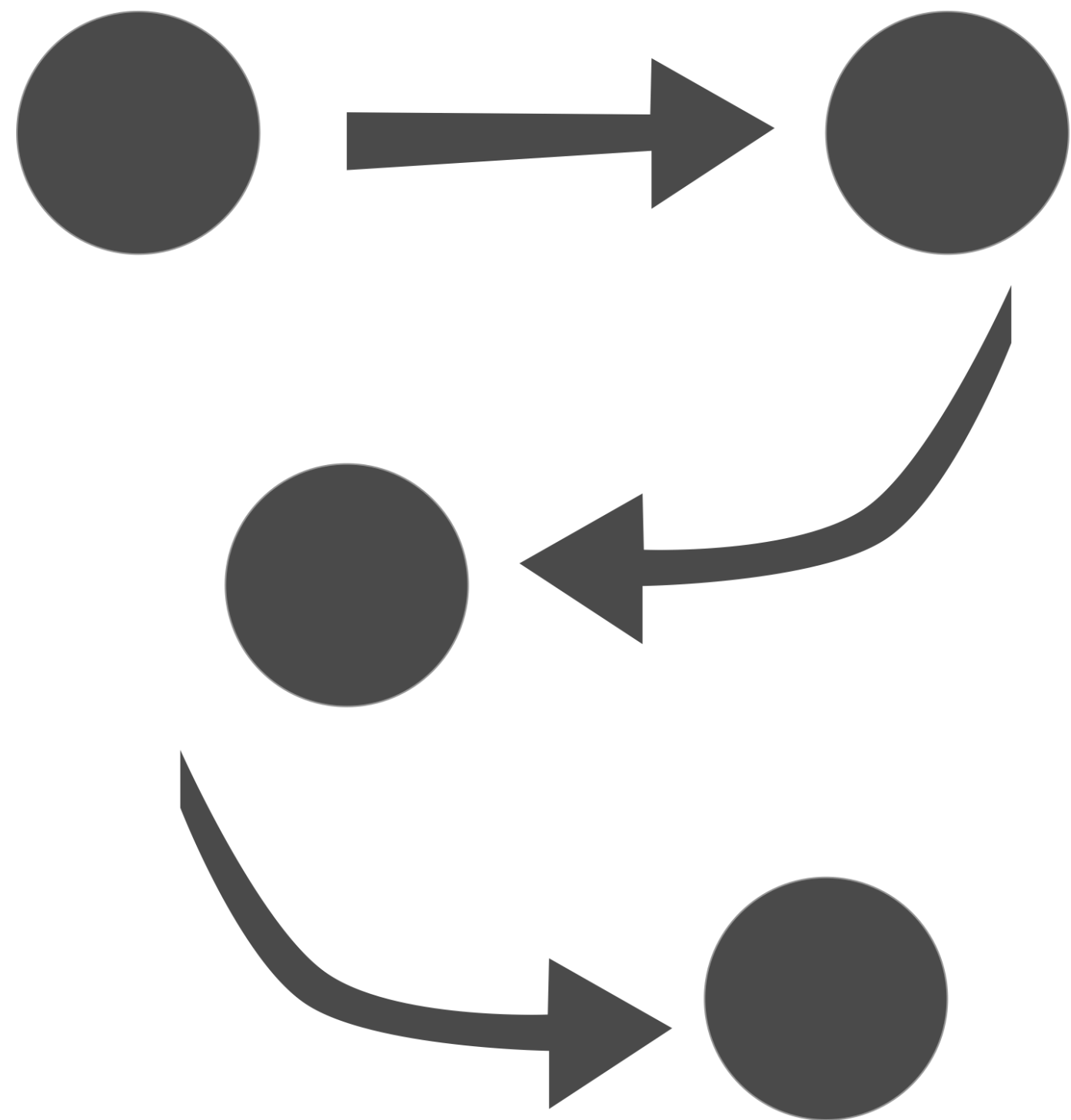
BBDD Relacionales

Pros y Contras



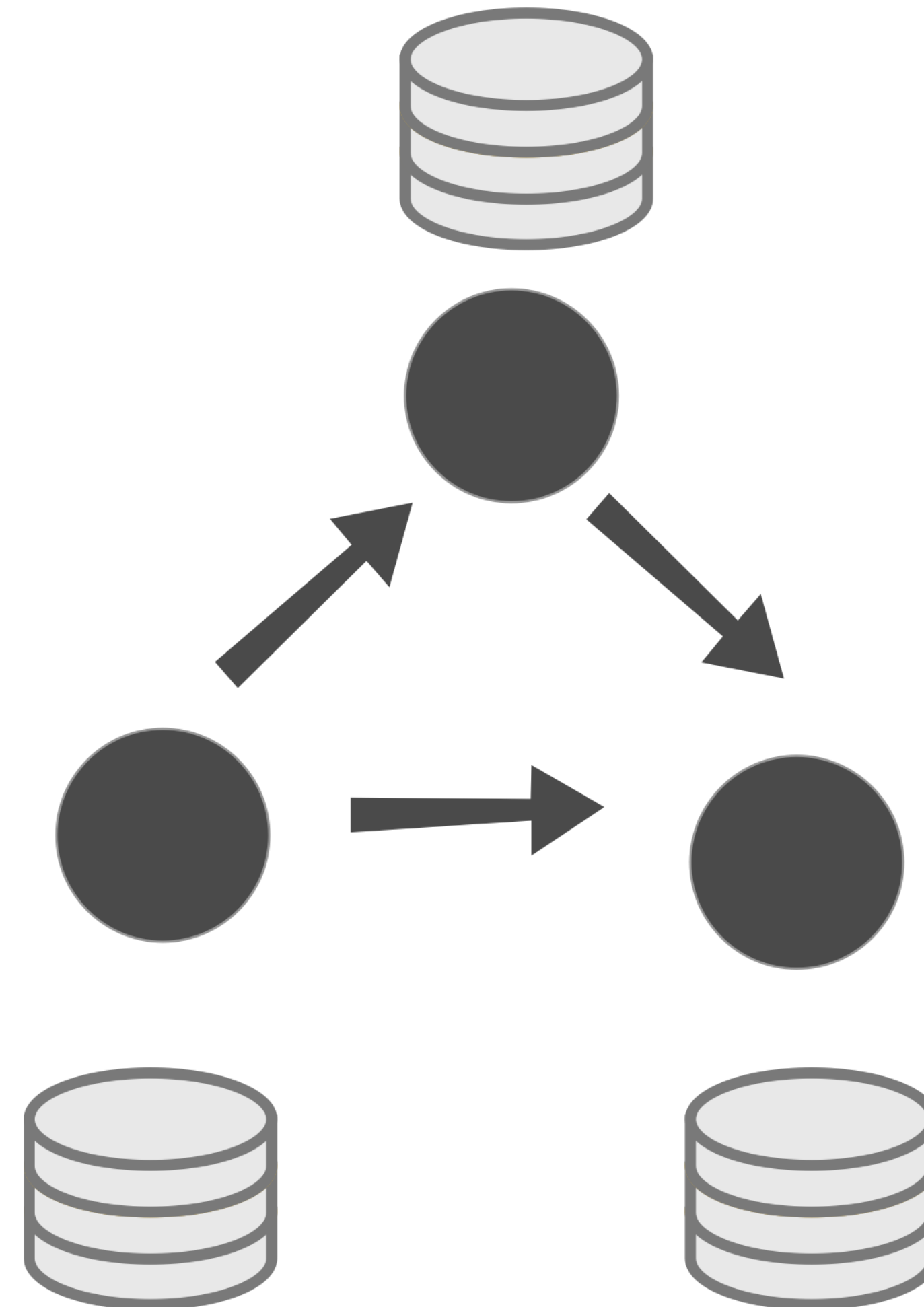
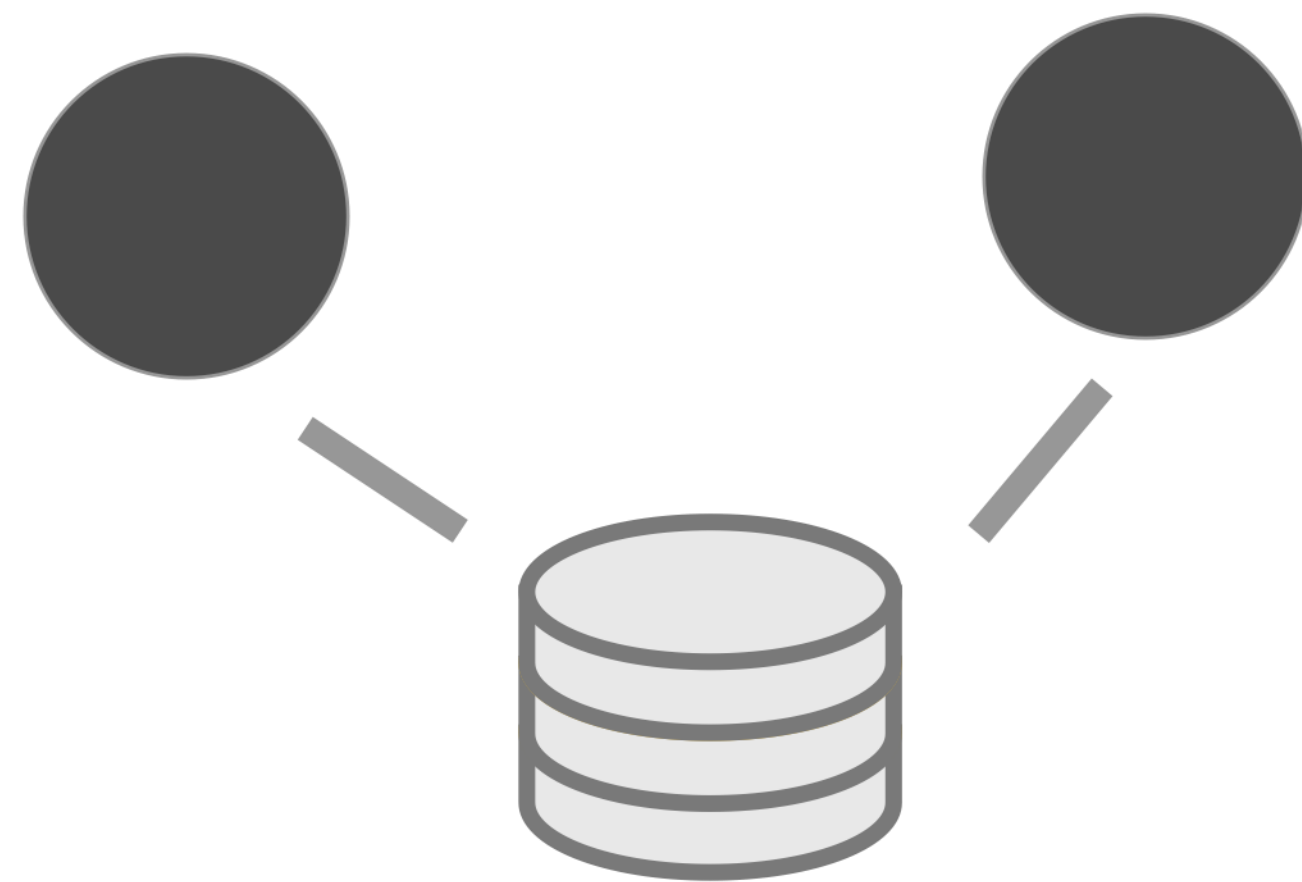
BBDD Relacionales

Diferencia de Impedancia



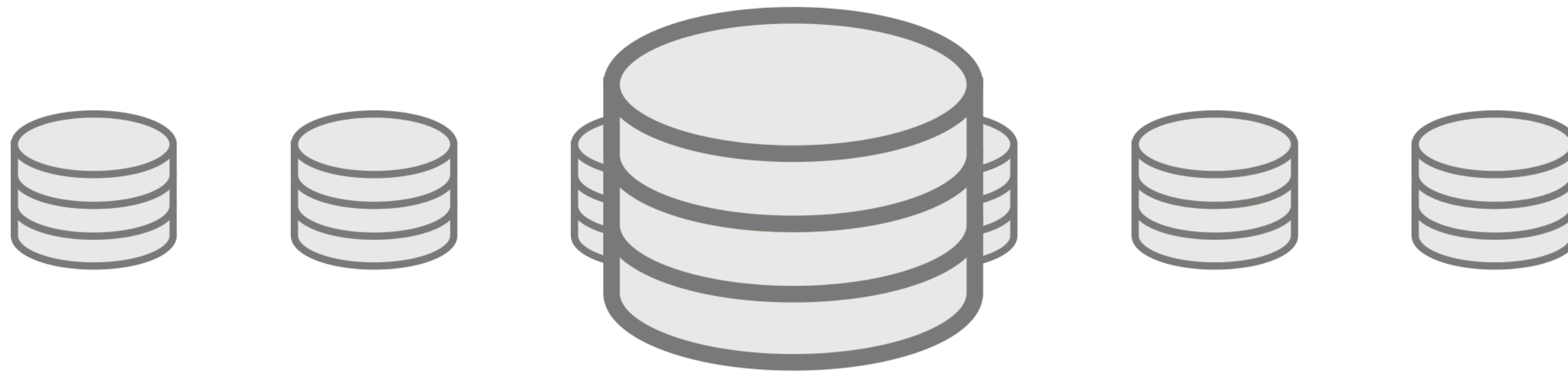
BBDD Relacionales

Integración



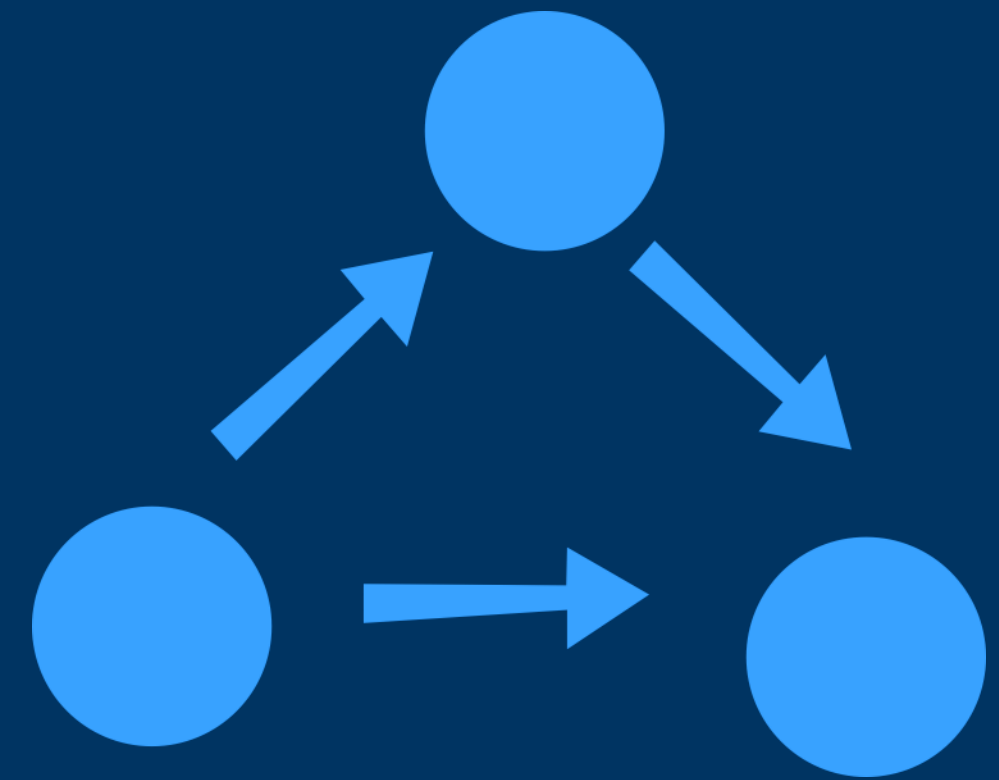
BBDD Relacionales

Volumen de Datos



BBDDs NoSQL

Nuevas generaciones



#NoSQL

Bases de Datos No-Relacionales



3124

PERFIL

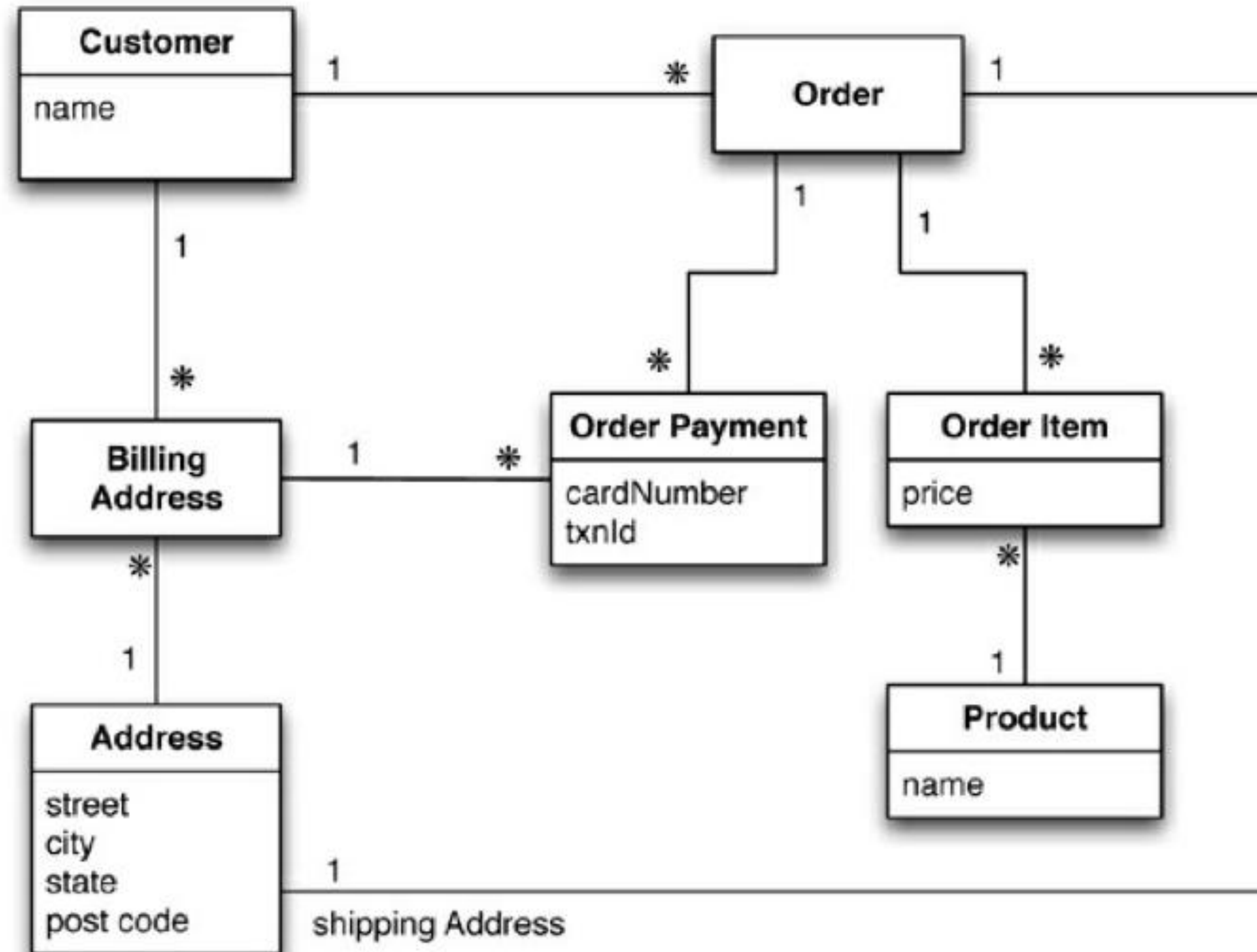
nombre	dirección
“Martin”	Calle 9 323

RESERVAS

res01	res02	res03
<...>	<...>	<...>

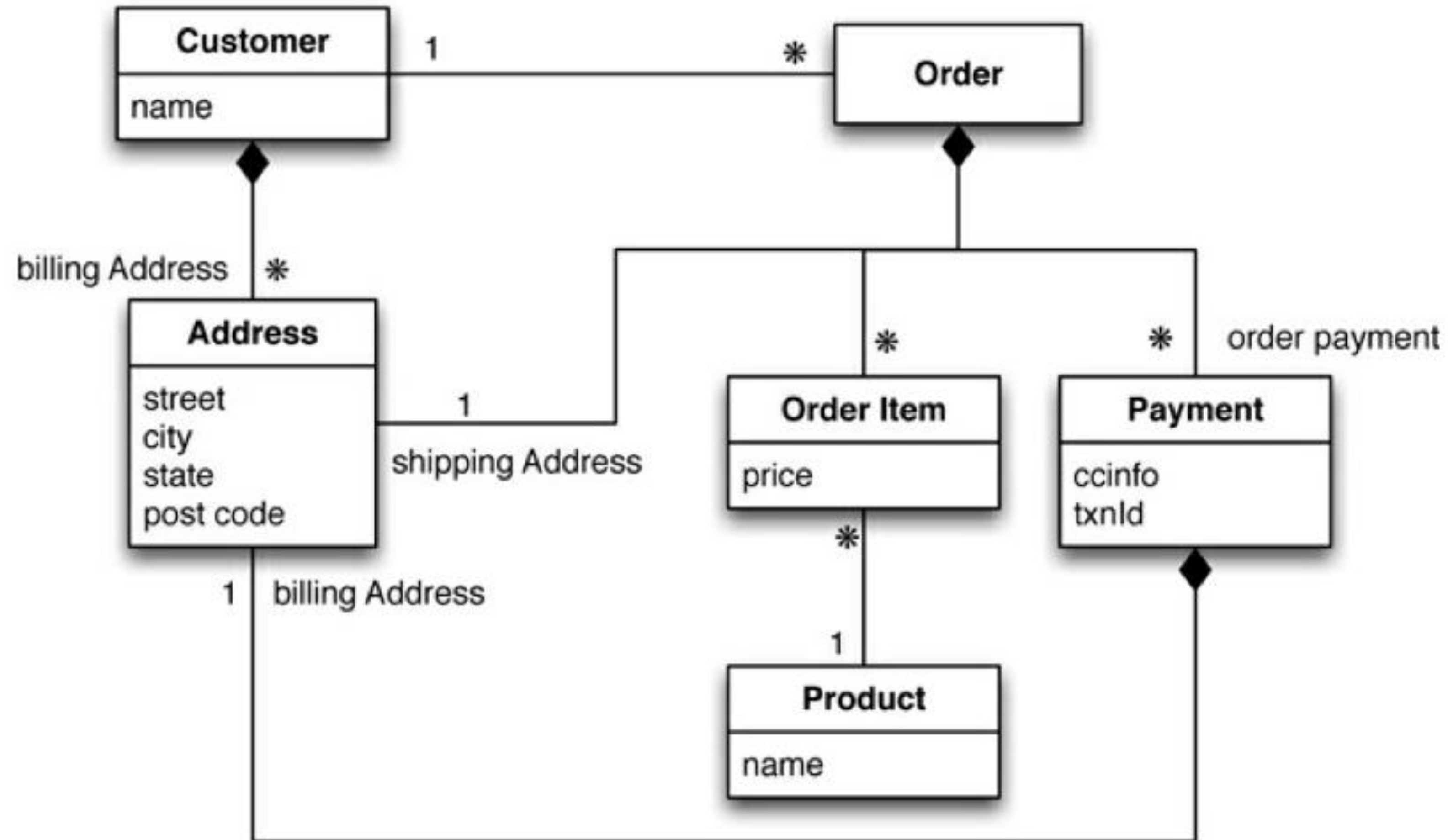
Agregaciones

No, esas no



Aggregaciones

Diseño



Aggregaciones

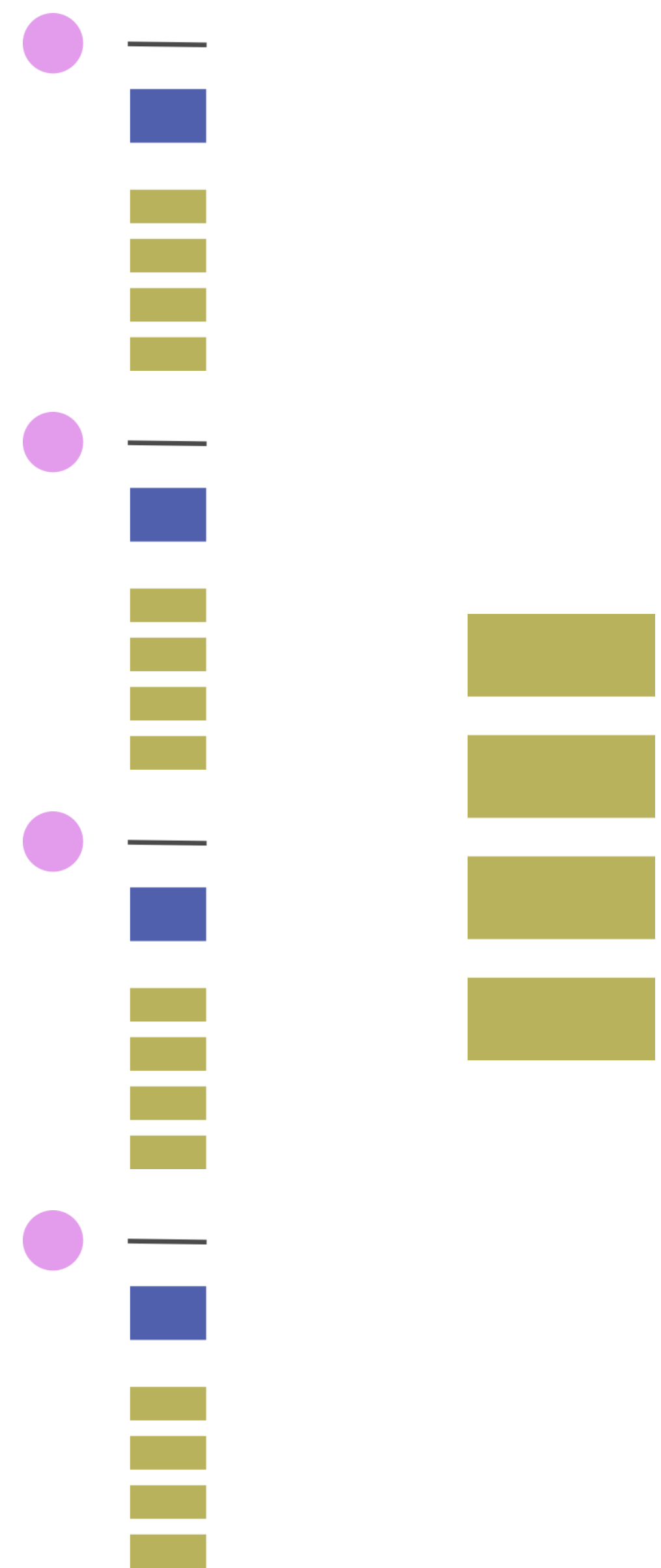
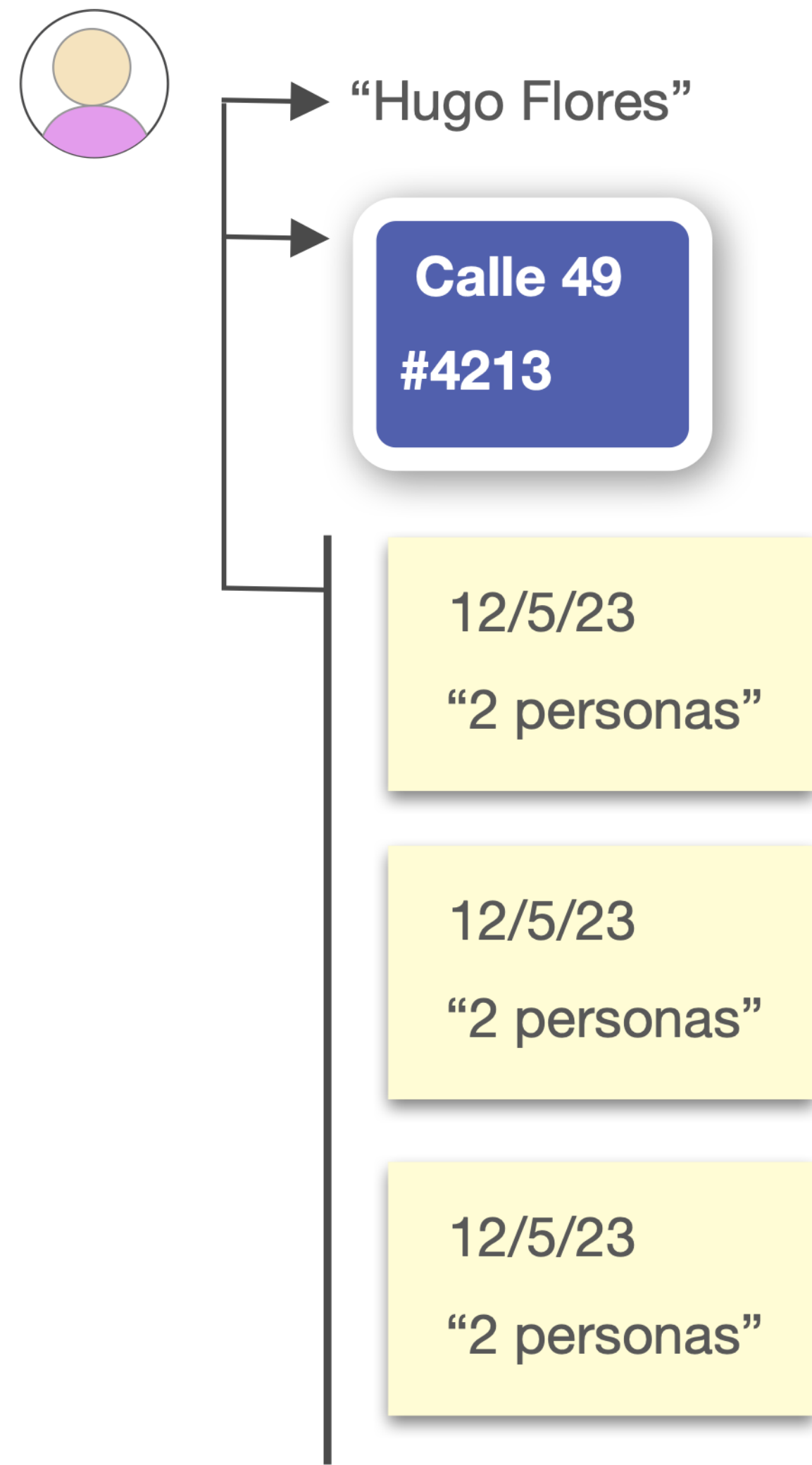
Diseño

```
// in customers
{
  "id":1,
  "name":"Martin",
  "billingAddress":[
    {"city":"Chicago"}
  ]
}
```

```
// in orders
{
  "id":99,
  "customerId":1,
  "orderItems":[
    {
      "productId":27,
      "price": 32.45,
      "productName": "NoSQL Distilled"
    } ],
  "shippingAddress":[{"city":"Chicago"}]
  "orderPayment":[
    {
      "ccinfo":"1000-1000-1000-1000",
      "txnId":"abelif879rft",
      "billingAddress": {"city": "Chicago"}
    }
  ],
}
```

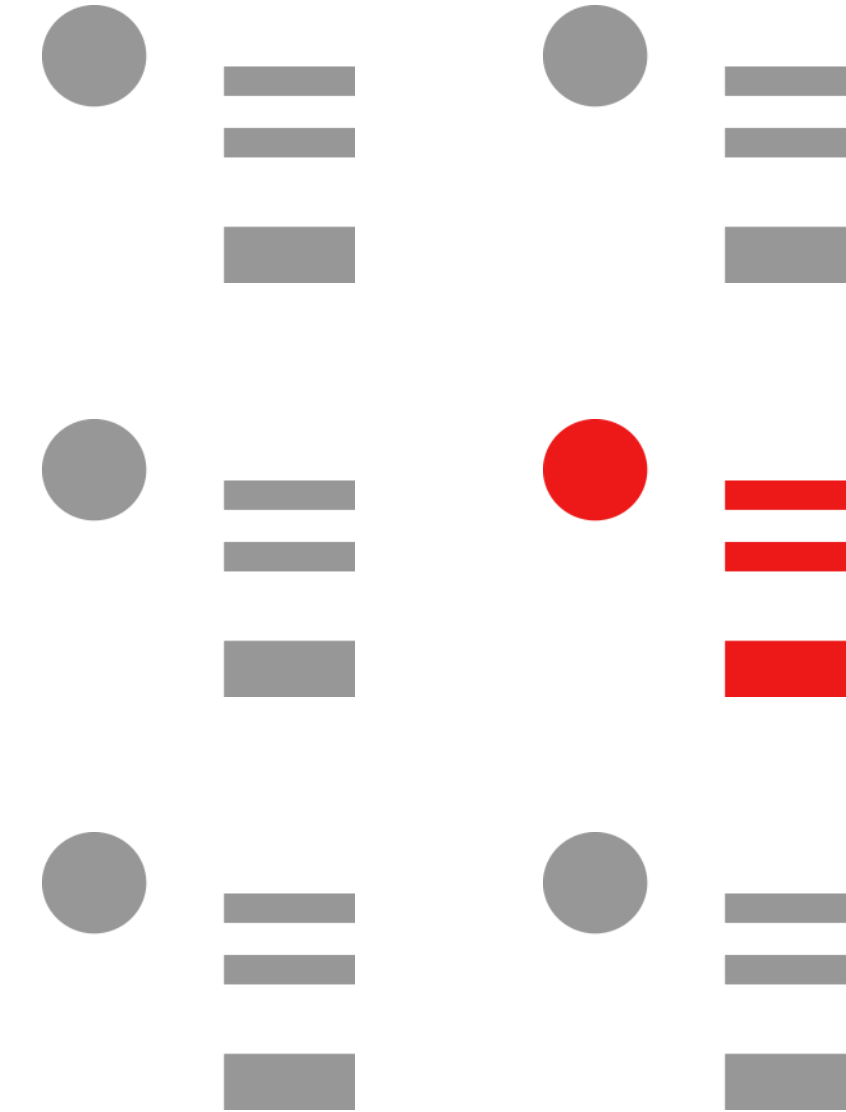
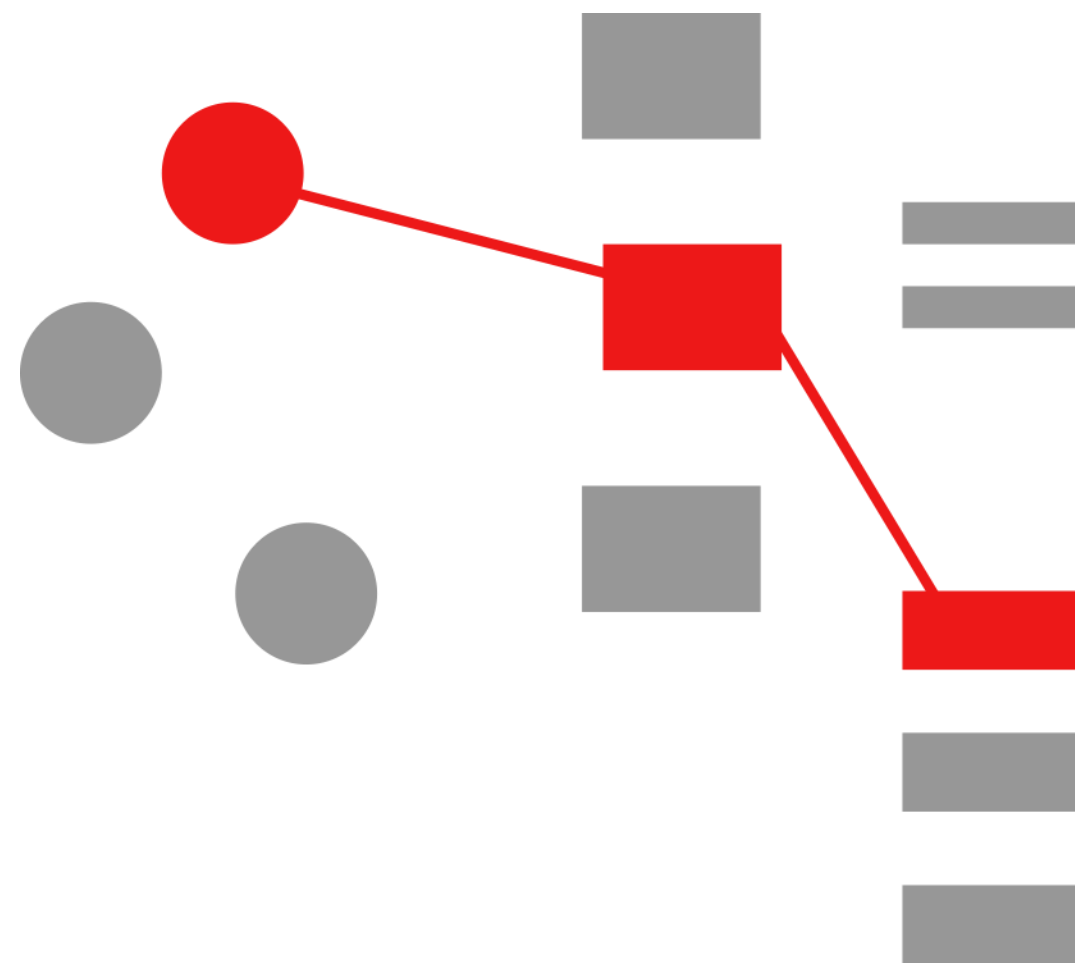

Agregaciones

Consideraciones



Agregaciones

Transacciones

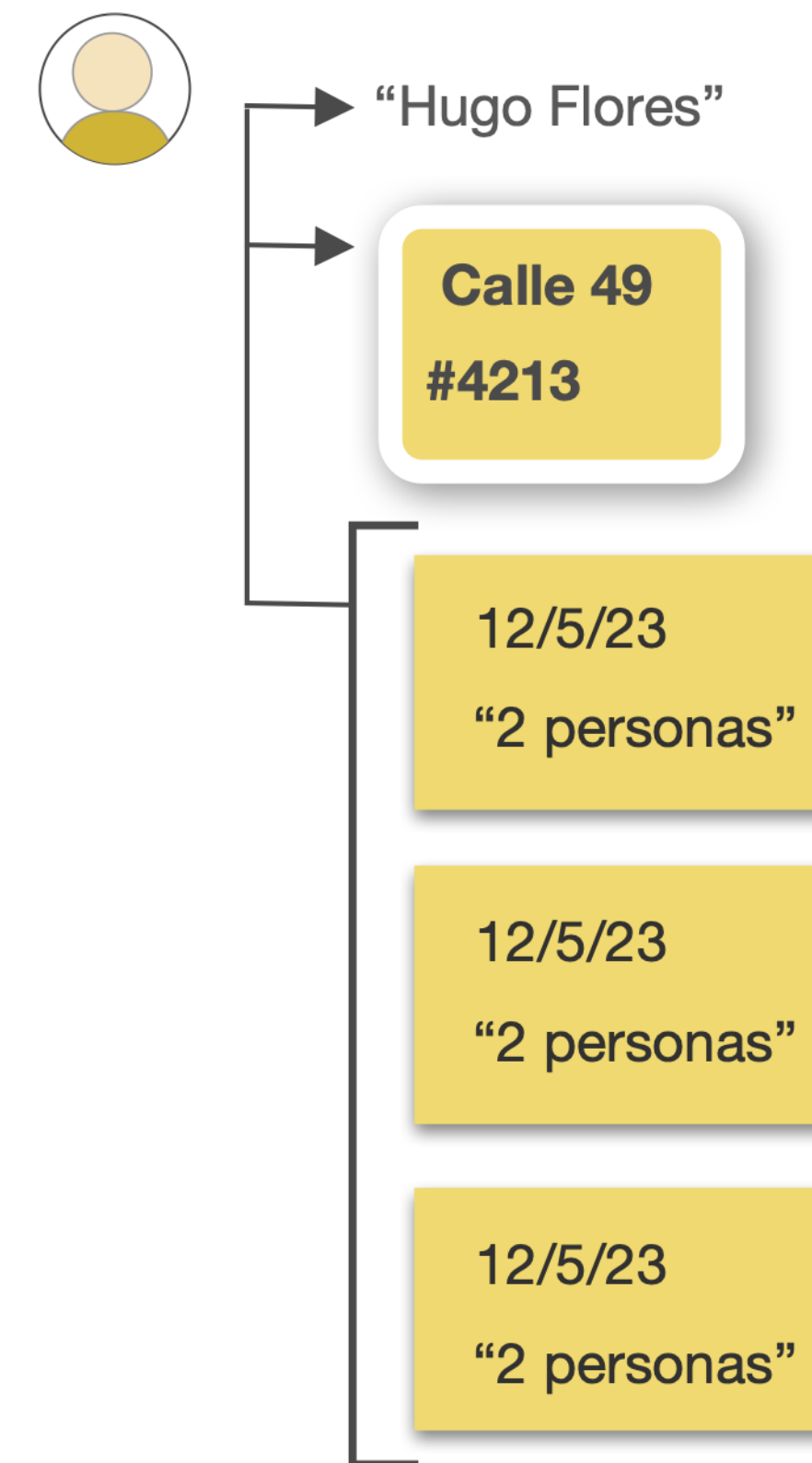


Familias NoSQL

Tipos de Modelos de Datos



Clave - Valor



Documentos

Familias NoSQL

Tipos de Modelos de Datos

PERFIL					RESERVAS		
id	no	nombre	dirección	s01	res01	res02	res03
3124	"Martín"	"Martín"	calle 123	323<...>	<<...>	<<...>	<...>

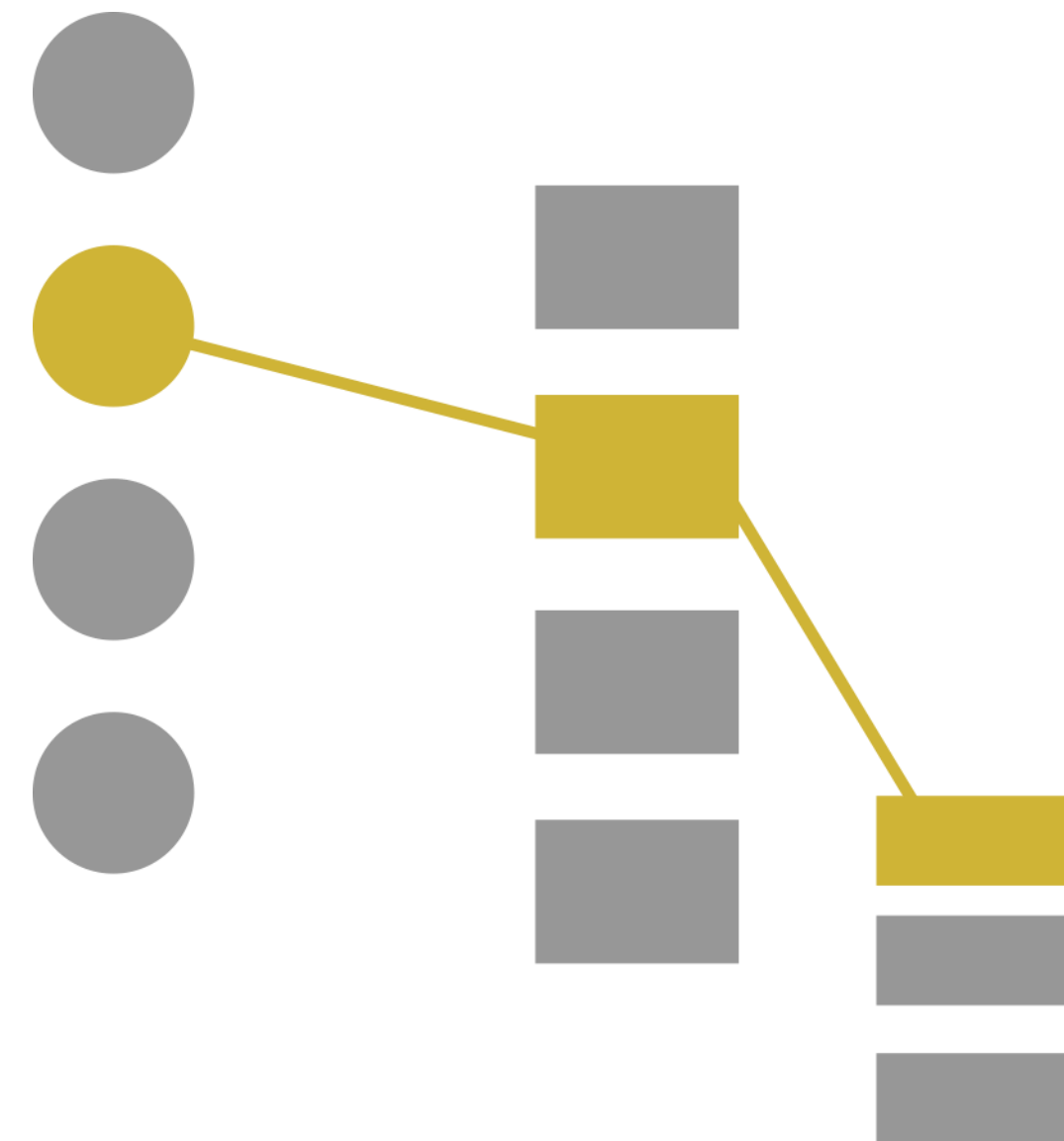
Columna

NoSQL

Manejo de Relaciones

Relaciones

NoSQL vs. RDMS



Relaciones

NoSQL vs. RDMS



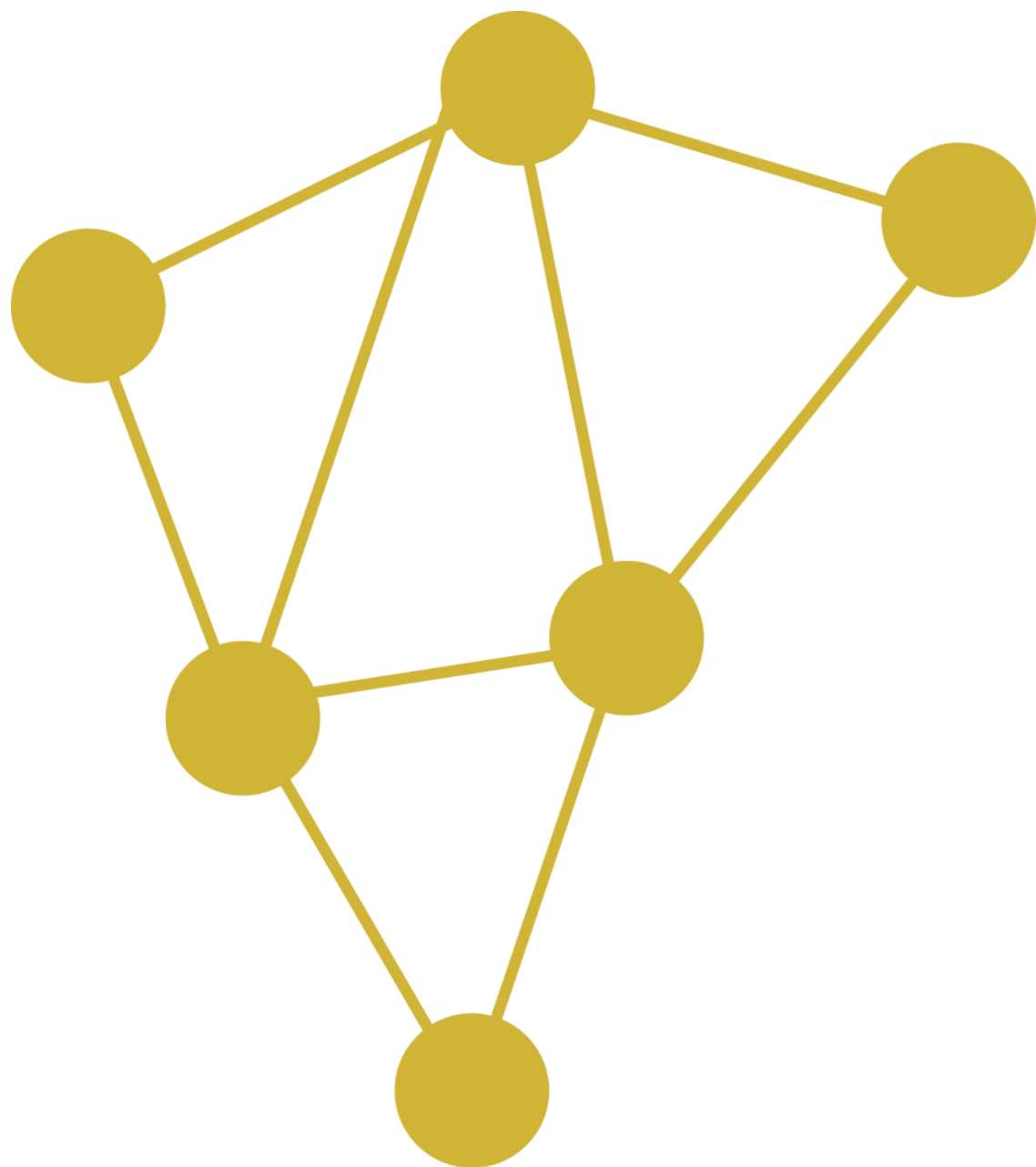
Duis mollis, est non commodo luctus, nisi erat porttitor ligula, eget lacinia odio sem nec elit. Donec id elit non mi porta gravida at eget metus. Cum sociis natoque penatibus et



2389123812301923
3192093120931203192391
9310230192391023
19239012390



{name: "John Doe"
dob: 12/12/1990
interests: ["Fishing", "Music"]
}





Hugo Flores

Calle 49
#4213

12/5/23
"2 personas"

12/5/23
"2 personas"

12/5/23
"2 personas"



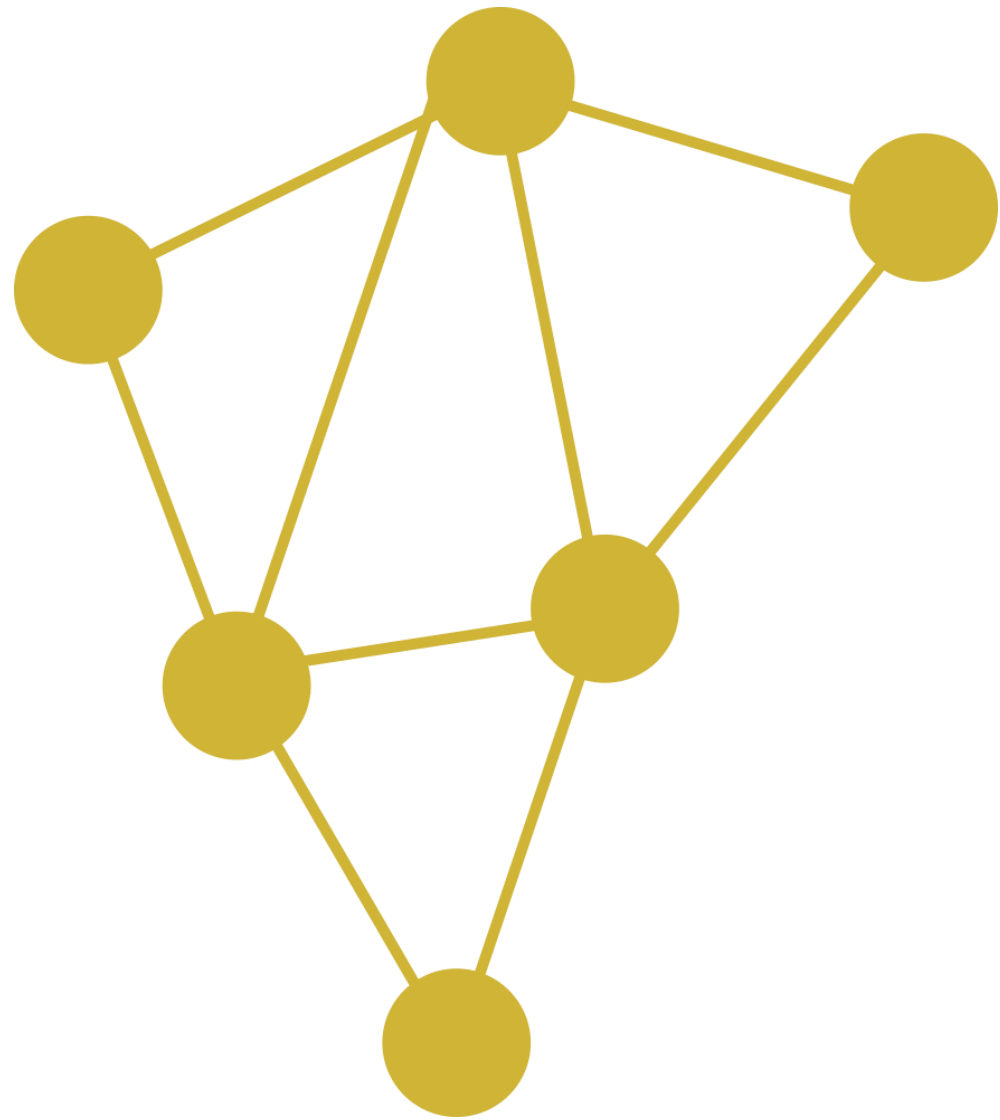
3124

PERFIL	
nombre	dirección
"Martin"	Calle 9 323

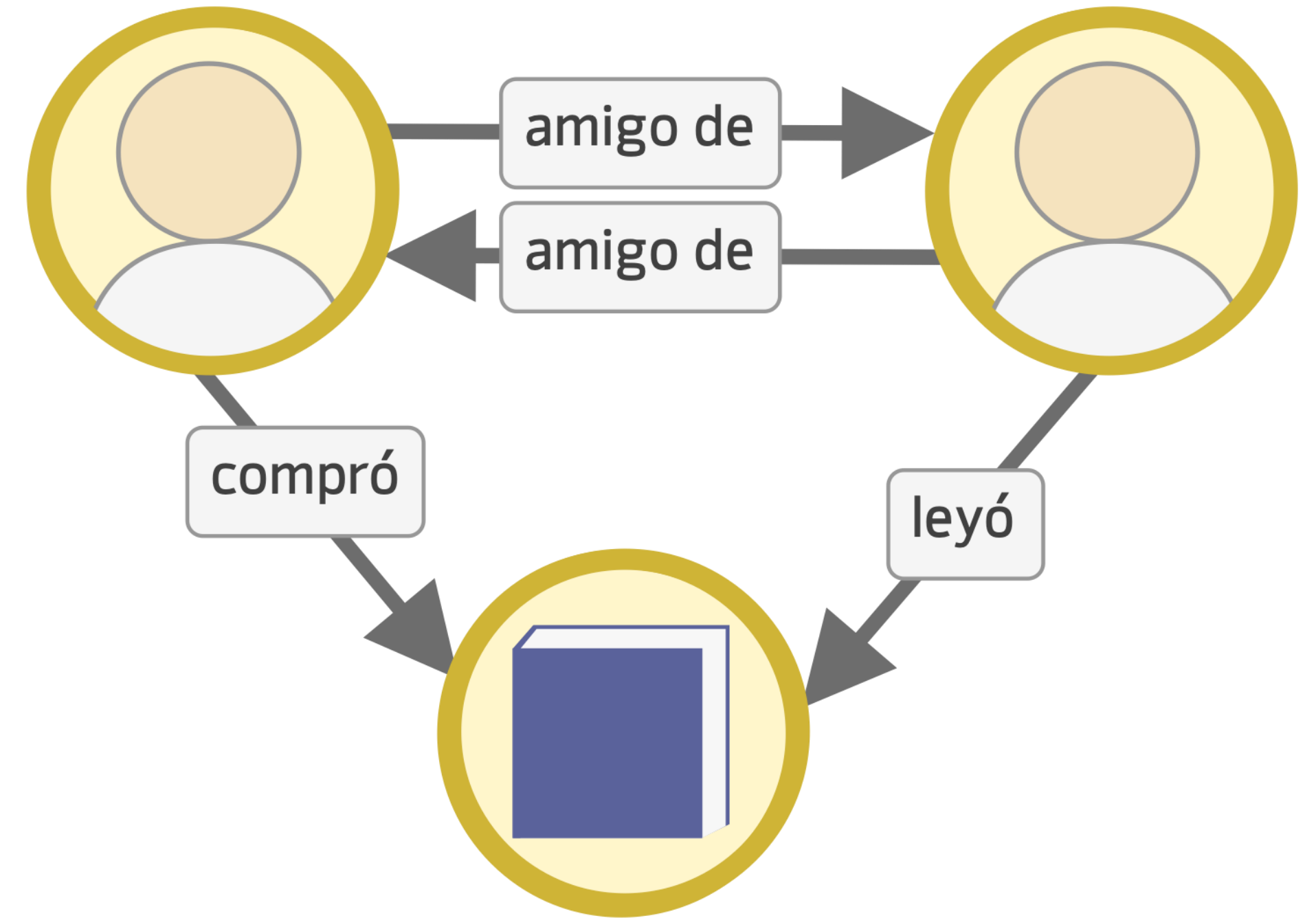
RESERVAS		
res01	res02	res03
<...>	<...>	<...>

BBDD Orientadas a Grafos

Ejemplos



 neo4j



Pathfinding
and Search



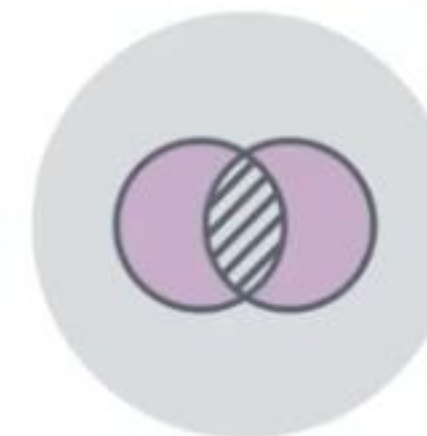
Centrality



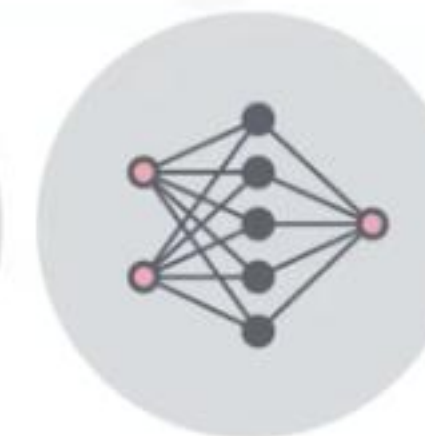
Community
Detection



Heuristic
Link Prediction

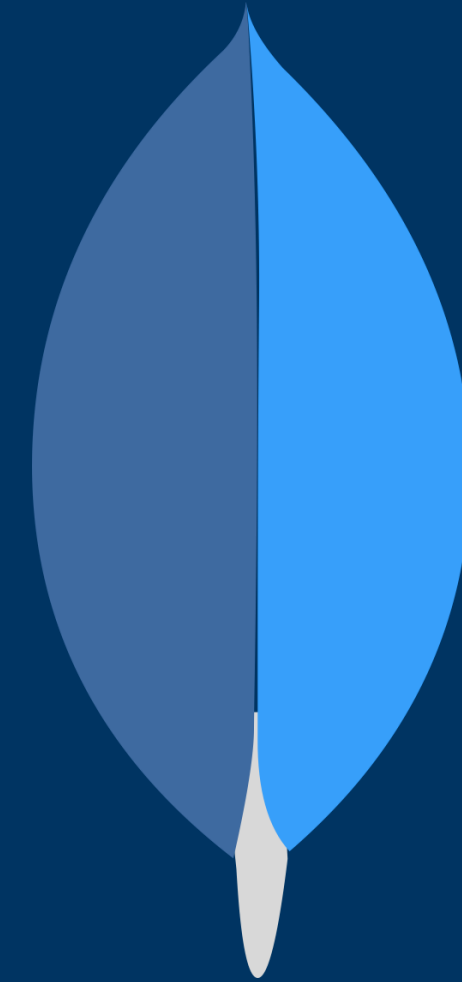


Similarity



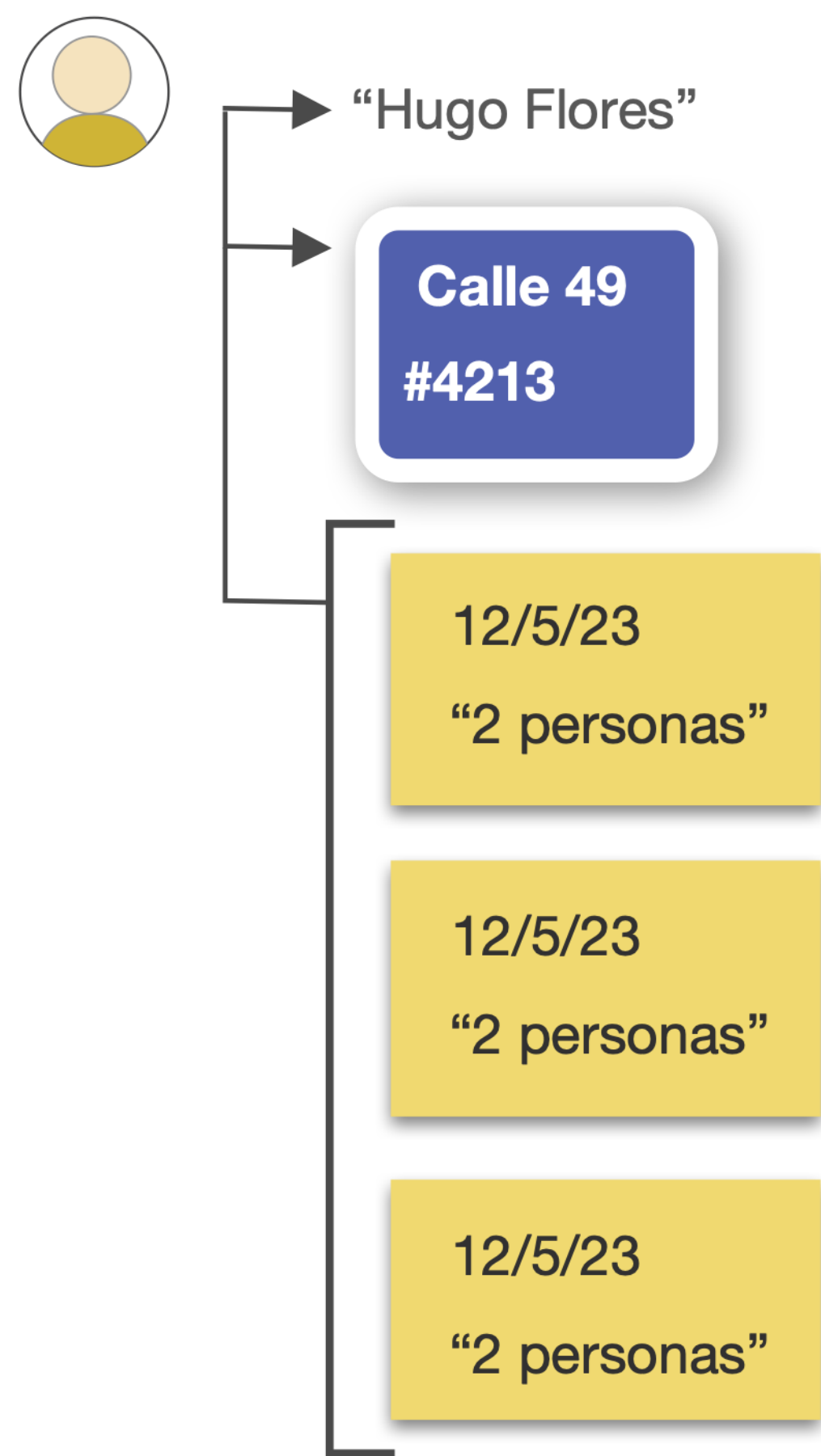
Embeddings

MongoDB



MongoDB

Características



```
{
  "id":99,
  "nombre": "Hugo Pérez",
  "dirección": {
    "calle": "Calle 49",
    "numero": 4213
  }
  "reservas":[
    {"fecha": 12/3,
    "detalle": "2 p"},
    {"fecha": 12/5,
    "detalle": "2 p"},
    {"fecha": 15/7,
    "detalle": "2 p"}
  ]
}
```

```
\x16\x00\x00\x00
\x02
hello\x00
\x06\x00\x00\x00world\x00
\x00
```

Almacenamiento

MongoDB



MongoDB

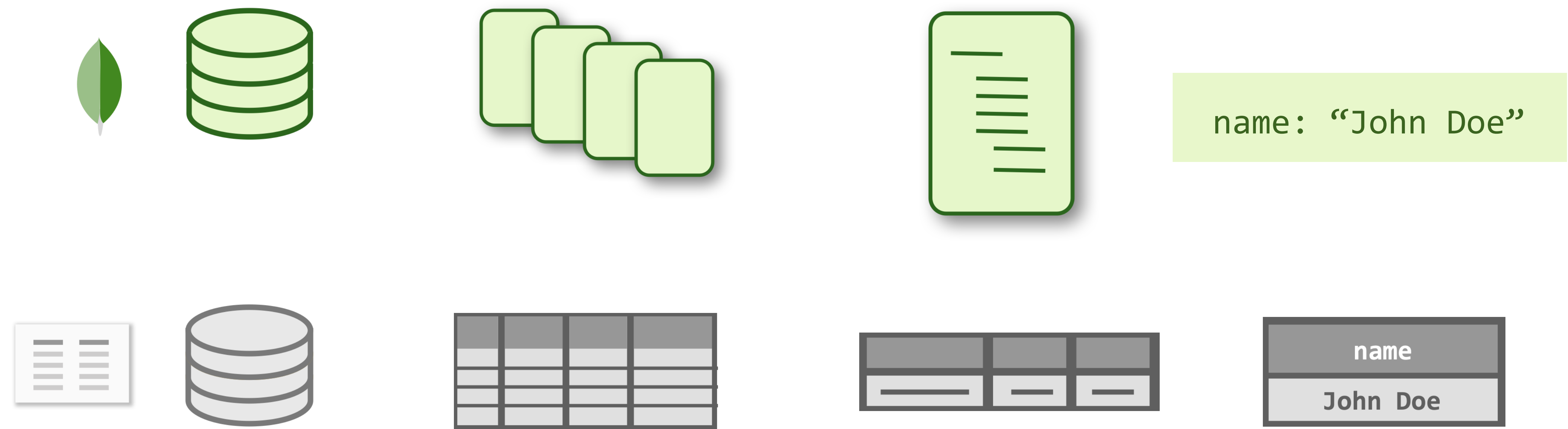
“Schemaless”

```
{
  "id":99,
  "nombre": "Hugo Pérez",
  "dirección": {
    "calle": "Calle 49",
    "numero": 4213
  }
  "reservas":[
    {"fecha": 12/3,
    "detalle": "2 p"},
    {"fecha": 12/5,
    "detalle": "2 p"},
    {"fecha": 15/7,
    "detalle": "2 p"}
  ]
}
```

```
{
  "id":99,
  "nombre": "Ana Lamas",
  "dirección": {
    "calle": "Calle 4",
    "numero": 323,
    "detalle": esq 40
  }
}
```

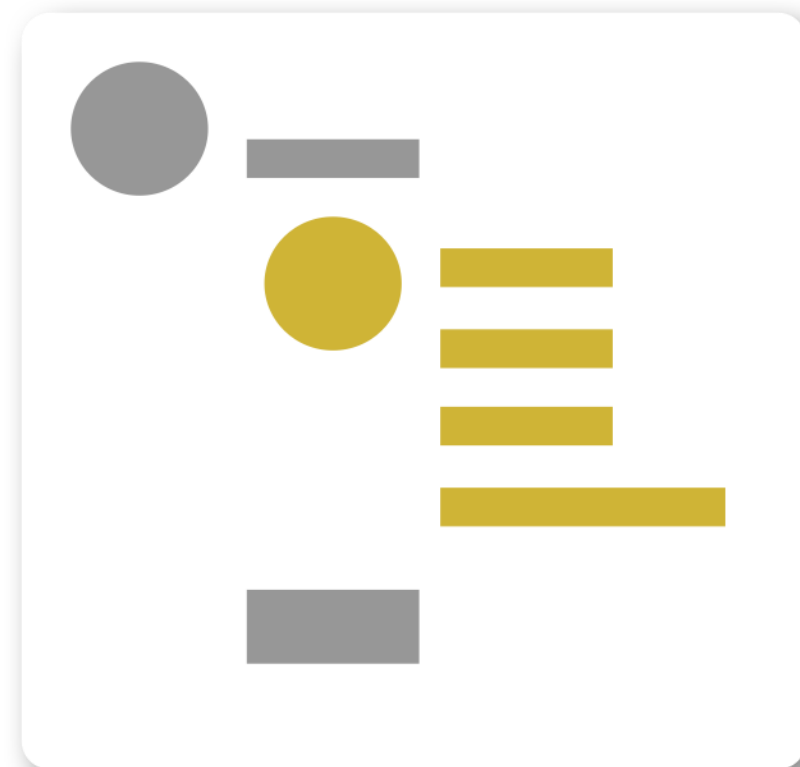
MongoDB

Estructura vs. RDBMS

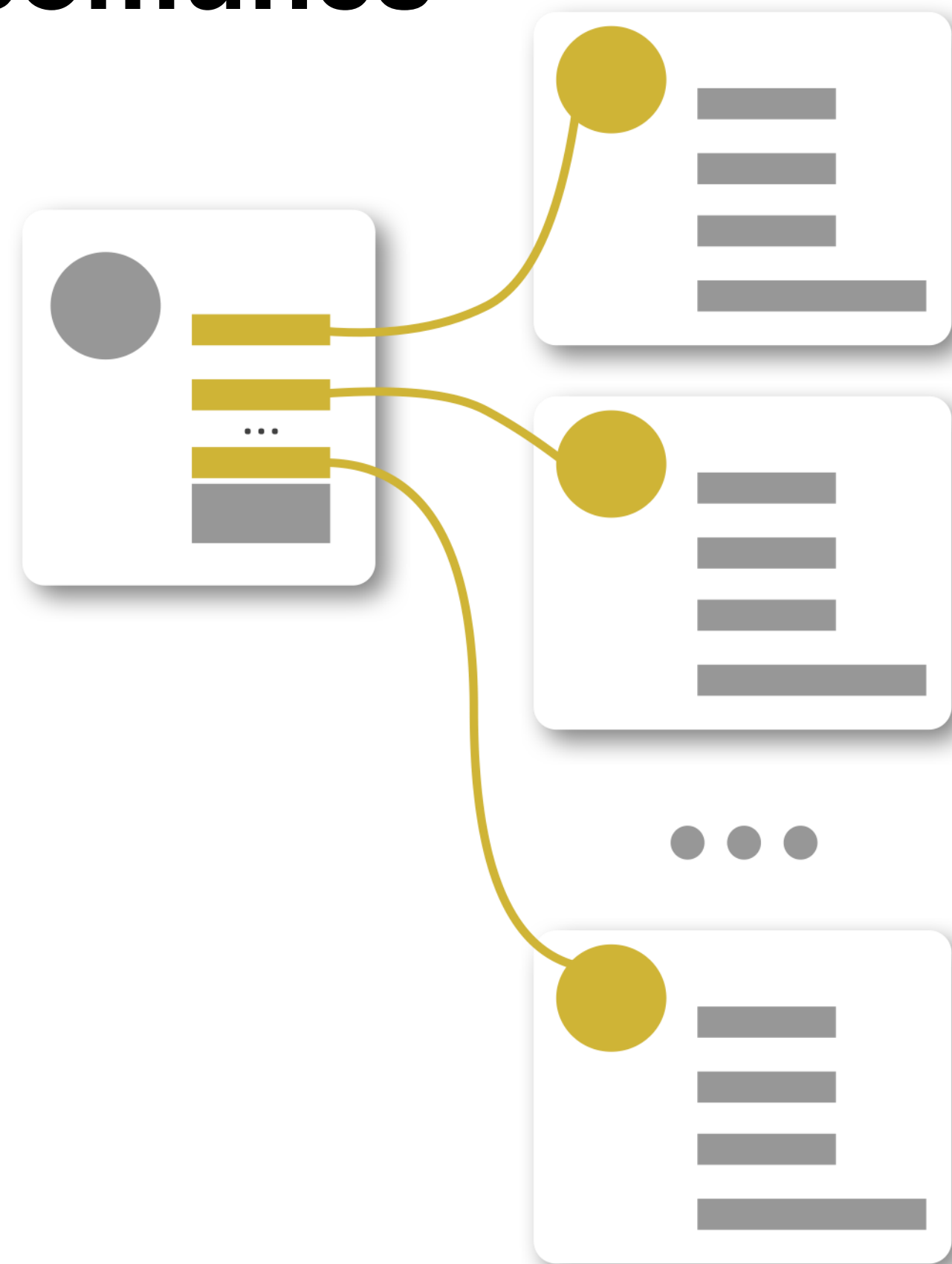


MongoDB

Relaciones - Prácticas comunes

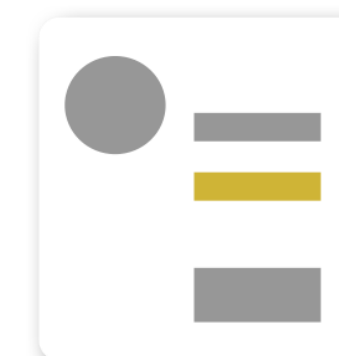


1 a Varios



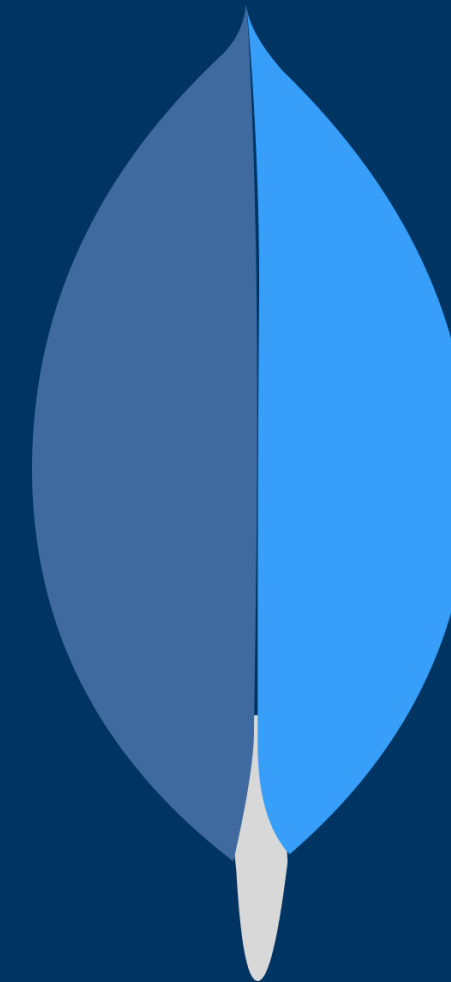
1 a Muchos

1 a Millones



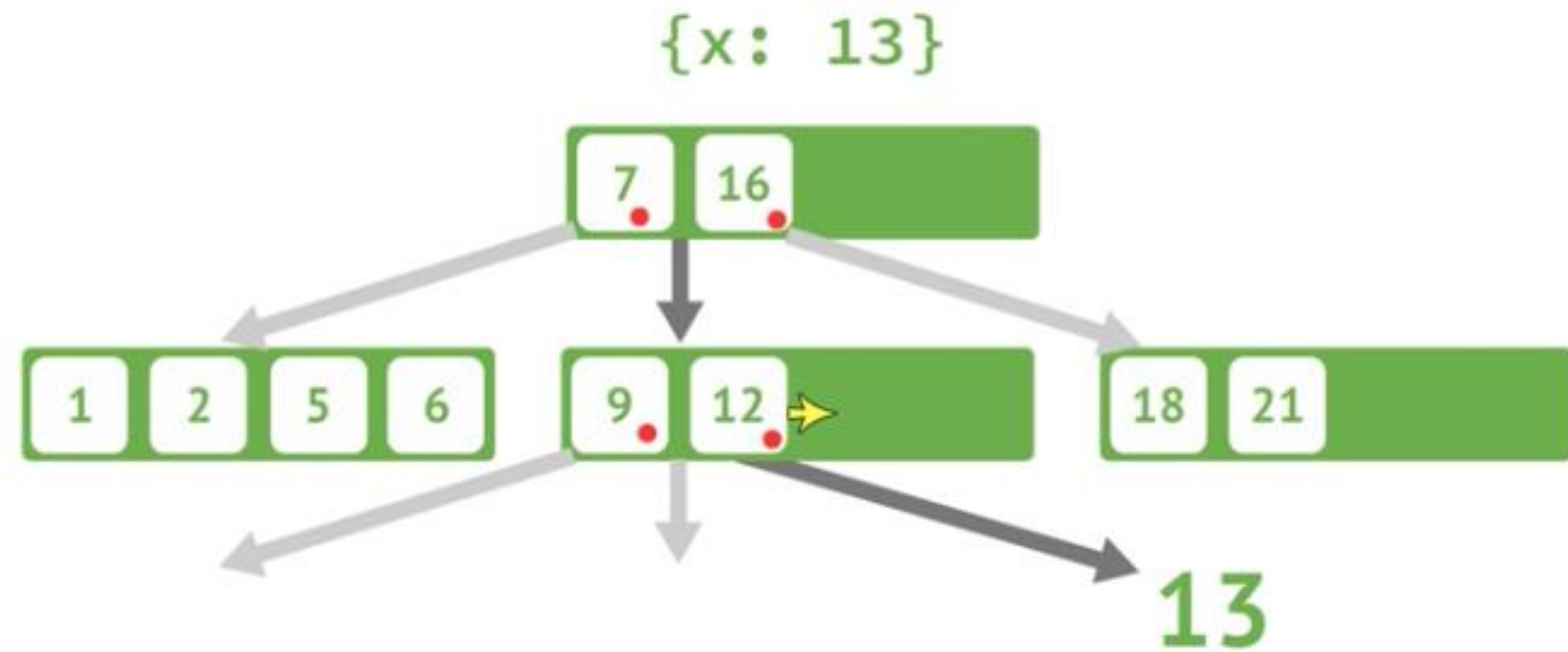
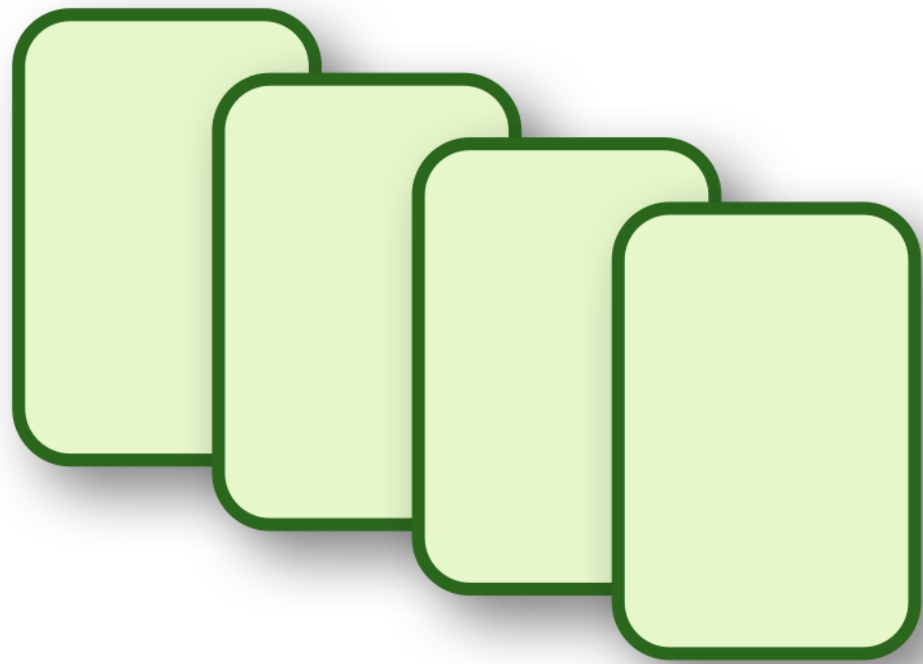
Demo

Operaciones CRUD en MongoDB



Indices

En MongoDB



Indices

En MongoDB

- Single field
 - `db.collection.createIndex( <key and index type specification>, <options> )`
 - `db.records.createIndex( { score: 1 } )`
 - `db.records.createIndex( { "location.state": 1 } )`

Indices

En MongoDB

- Compound field
 - `db.collection.createIndex( { <field1>: <type>, <field2>: <type2>, ... } )`
 - `db.products.createIndex( { "item": 1, "stock": 1 } )`

Indices

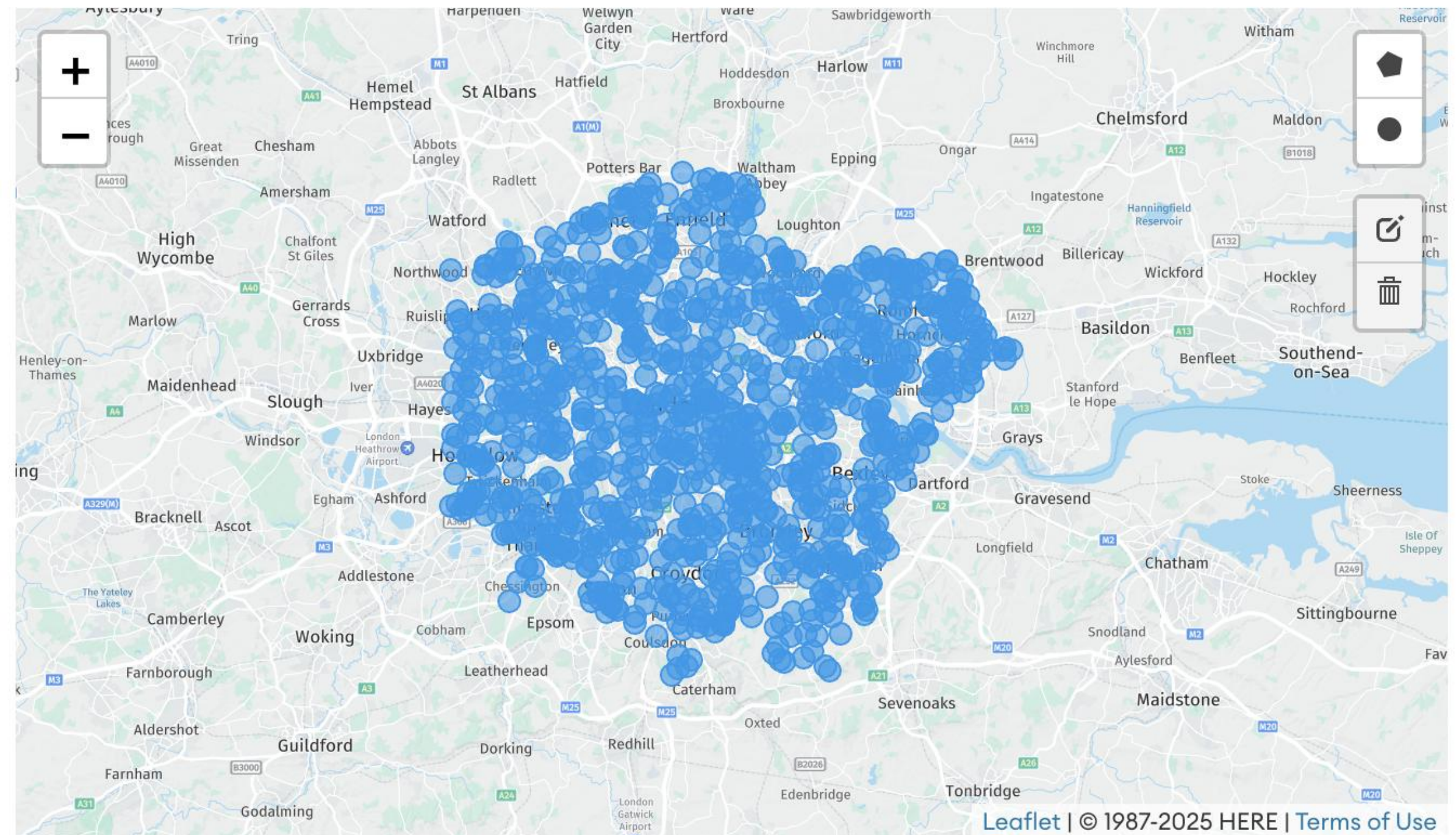
En MongoDB

- Geospatial field
 - `db.collection.createIndex( { <location field> : "2dsphere" } )`
 - `db.places.createIndex( { loc : "2dsphere" } )`
 - `db.places.createIndex( { loc : "2dsphere" , category : -1, name: 1 } )`

Indices

En MongoDB

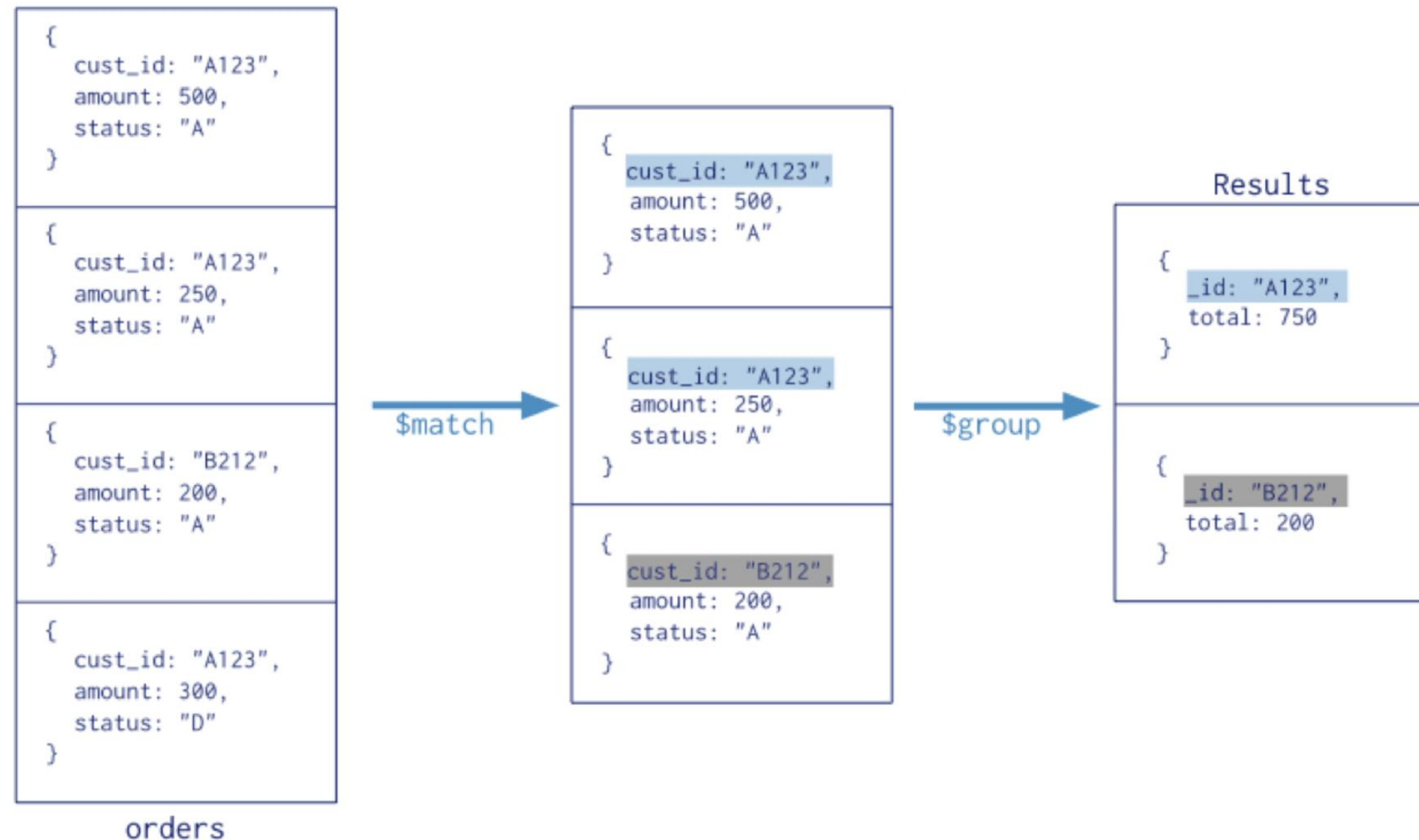
```
db.apartments.find({  
  location: { $geoWithin: { $geometry : london } }  
})
```



Aggregation Framework

Consultas complejas en MongoDB

Collection
↓
db.orders.aggregate([
 \$match stage → { \$match: { status: "A" } },
 \$group stage → { \$group: { _id: "\$cust_id", total: { \$sum: "\$amount" } } }
])



Aggregation Framework

Consultas complejas en MongoDB

